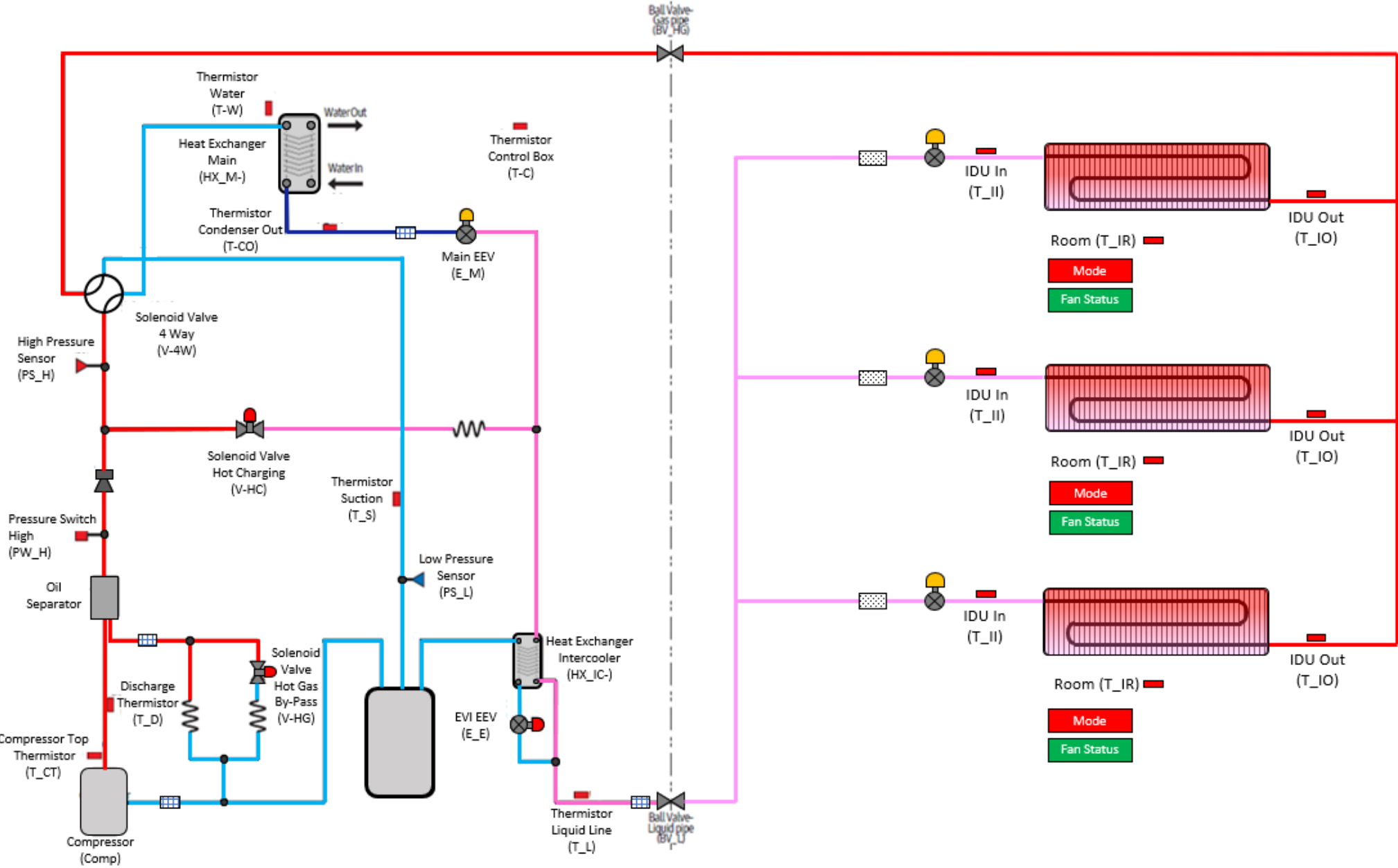




DVM S Single Phase Water  
Heat Pump Operation Evaluation  
Heating Mode



Comments:

| Outdoor Unit      |                                                 |         |
|-------------------|-------------------------------------------------|---------|
| Component         | Target                                          | Reading |
| Water Out Temp.   |                                                 |         |
| Compressor Hz.    |                                                 |         |
| High pressure     | Based on water temp                             |         |
| H.P Sat. Temp     |                                                 |         |
| Discharge Temp.   | 150 <sup>0</sup> -212 <sup>0</sup>              |         |
| IPM Temp,         | Below 194 <sup>0</sup>                          |         |
| Low pressure      | 114 Psig.                                       |         |
| L.P. Sat. Temp    | 38 <sup>0</sup>                                 |         |
| Suction 1 Temp.   |                                                 |         |
| Suction 2 Temp    |                                                 |         |
| S.H.              | 3.6 <sup>0</sup> or 9 <sup>0</sup>              |         |
| Min EEV Position  | Modulating                                      |         |
| Cond. Out Temp.   | Water Out – (2 <sup>0</sup> – 35 <sup>0</sup> ) |         |
| Liquid Line Temp. |                                                 |         |
| Sub-Cooling       | 9 <sup>0</sup> or 36 <sup>0</sup>               |         |
| Indoor Units      |                                                 |         |
| Component         | Target                                          | Reading |
| IDU 1 EEV         |                                                 |         |
| IDU 1 SC          | 9 <sup>0</sup>                                  |         |
| IDU 1 T PI        |                                                 |         |
| IDU 1 T PO        |                                                 |         |
| IDU 2 EEV         |                                                 |         |
| IDU 2 SC          | 9 <sup>0</sup>                                  |         |
| IDU 2 T PI        |                                                 |         |
| IDU 2 T PO        |                                                 |         |
| IDU 3 EEV         |                                                 |         |
| IDU 3 SC          | 9 <sup>0</sup>                                  |         |
| IDU 3 T PI        |                                                 |         |
| IDU 3 T PO        |                                                 |         |

Directions:

Enter the data in the spaces provided. The reading will populate in the refrigerant diagram. Space temperature is entered at the IDU. Use the drop down boxes to identify the state of the component. Use the operating data and the target operation data on the next page to troubleshoot the system.



DVM S 3 Phase Water  
Heat Pump Target Operation

| Water Source Outdoor Unit             |              |           |                                                             |                                                                                                         |
|---------------------------------------|--------------|-----------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Component                             | Abbreviation |           | Heating Mode                                                | Comments                                                                                                |
| Compressor                            | Comp         | Operation | Target High pressure control (default: 406 psig)            | Primary function is to maintain head pressure (range 356-455psig)                                       |
|                                       |              | Range     | 20~160Hz                                                    |                                                                                                         |
| Main EEV                              | E_M          | Operation | HR - Target SH = Water temperature - 5.4°F                  | Valve modulates to maintain Evaporator Superheat                                                        |
|                                       |              | Range     | 125 - 2000 Steps                                            | Suction Superheat 1.8°F - 12.6°F                                                                        |
| EVI EEV                               | E_EVI        | Operation | On (Open)                                                   | By-Pass Mode 18°F Superheat Between pipe in and pipe out                                                |
|                                       |              | Range     | 0-480step                                                   | EVI Vapor Injection Mode 3.6°F Superheat Between pipe in and pipe out                                   |
| 4 Way Valve                           | V_4W         | Operation | On (Energized)                                              |                                                                                                         |
| EVI By-Pass Solenoid Valve            | V_EB         | Operation | On (Open): *T_W > 50.4 or 50Hz↓                             | By-pass mode, Discharge Temperature control                                                             |
|                                       |              |           | Off (Close): T_W < 45°F ↓ and 65Hz↑                         | Low Ambient heating Operation Vapor Injection circuit                                                   |
| EVI Solenoid Valve                    | V_EVI        | Operation | On (Open)                                                   |                                                                                                         |
| Hot Gas By-Pass                       | V_HG         | Operation | Off (Closed)                                                |                                                                                                         |
| Accumulator Oil Return Solenoid Valve | V_AR         | Operation | On (Open)                                                   | <b>Off</b> during start up if DSH is < 18°F. <b>Off</b> for 2 min. if DSH is < 18°F On when DCSH > 27°F |
| Hot Gas By-Pass 2 (HR Only)           | V_HG2        | Operation | Off Closed                                                  |                                                                                                         |
| Liquid Line Solenoid Valve            | V_L          | Operation | On (Open)                                                   |                                                                                                         |
| Main Cooling Solenoid                 | V_MC         | Operation | Off (Closed)                                                |                                                                                                         |
| Hot Gas Charging Solenoid Valve       | V_HC         | Operation | Off (Closed)                                                |                                                                                                         |
| Condenser Out Thermistor              | T_CO         | Value     | Should be 2 degrees lower than the water out temperature    | Low Side saturation temperature is " <u>typically</u> " 2-10 degrees lower than water temperature       |
| Component                             | Abbreviation |           | Heating Mode                                                | Comments                                                                                                |
| Port Heating Solenoid                 | Heat         | Operation | On (Open)                                                   |                                                                                                         |
| Port Cooling Solenoid                 | Cool         | Operation | Off (Closed)                                                |                                                                                                         |
| Port EEV                              | EEV          | Operation | Closed 0 Steps                                              | Open when a zone satisfies                                                                              |
|                                       |              | Range     | 0 - 480 Steps                                               |                                                                                                         |
| Sub-Cooling EEV                       | SC_EEV       | Operation | Closed 0 Steps                                              | Active only in Heating Main and Cooling Main                                                            |
|                                       |              | Range     | 480 Steps                                                   |                                                                                                         |
| Sub-Cooler In Temperature             |              | Operation |                                                             |                                                                                                         |
| Sub-Cooler Out Temperature            |              | Operation | Close to Suction Header Temperature                         |                                                                                                         |
| Indoor Unit IDU                       |              |           |                                                             |                                                                                                         |
| Component                             | Abbreviation |           | Heating Mode                                                | Comments                                                                                                |
| Room Temperature                      | Room Temp.   | Operation | 61-86 Degrees                                               |                                                                                                         |
| Liquid Pipe Thermistor                | EVA In 1     | Operation | Liquid Pipe Temperature for sub-cooling Control             | Used to calculate Evaporator Sub-Cooling (91°F - 122°F)                                                 |
| Gas Pipe Thermistor                   | EVA Out 1    | Operation | Hot Gas Pipe Temperature should be close to discharge temp. | Used to evaluate the temperature of the hot gas coming into the coil (113°F - 194°F)                    |
| EEV                                   | EEV 1        | Operation | Modulates to maintain 9 degrees of sub-cooling              | Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.                                          |
|                                       |              | Range     | 500- 1500 Steps (80 - 120 Steps when zone is satisfied)     | Typical range between 500 and 1500 Steps                                                                |
| IDU Fan                               | Fan Speed    | Range     | Low , Medium, High, Auto                                    |                                                                                                         |
|                                       |              | Operation | Low Speed:  Ts - Ti  ≤ 1.8°F                                | TS - Set Temperature                                                                                    |
|                                       |              |           | Medium Speed: 1.8°F <  Ts - Ti  ≤ 5.4°F                     | Ti Indoor Temperature                                                                                   |
|                                       |              |           | High Speed 5.4°F <  Ts - Ti                                 |                                                                                                         |